



Data Mining:

3. Persiapan Data

Romi Satria Wahono

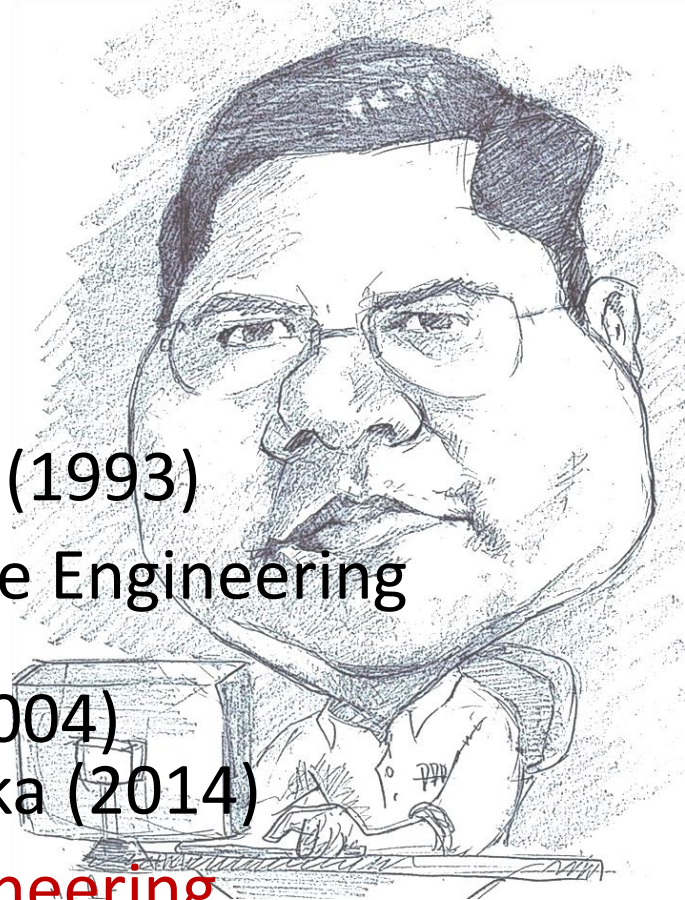
romi@romisatriawahono.net

<http://romisatriawahono.net/dm>

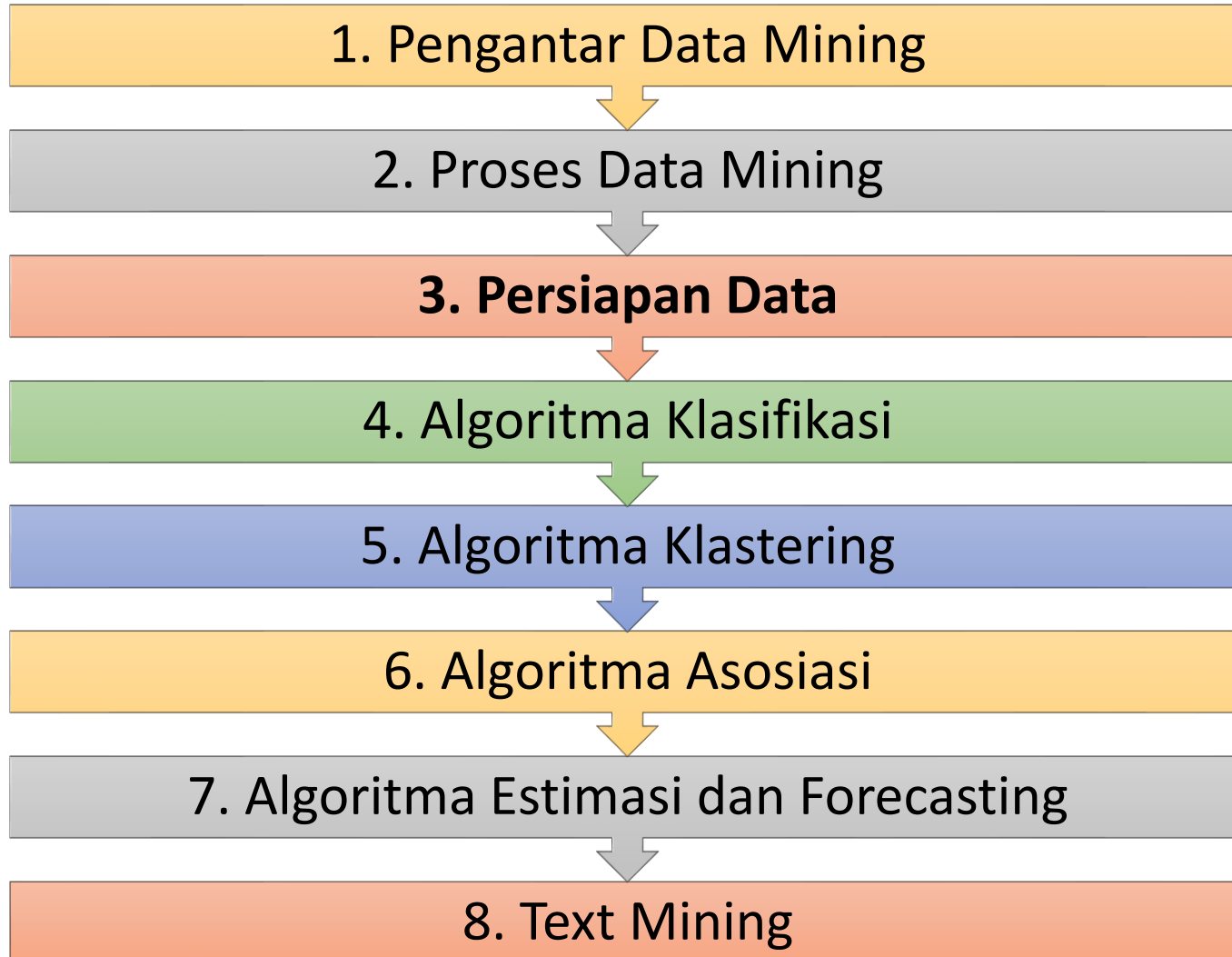
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Romi Satria Wahono

- **SD Sompok** Semarang (1987)
- **SMPN 8** Semarang (1990)
- **SMA Taruna Nusantara** Magelang (1993)
- **B.Eng, M.Eng** and **Ph.D** in Software Engineering from
Saitama University Japan (1994-2004)
Universiti Teknikal Malaysia Melaka (2014)
- Research Interests: **Software Engineering**,
Intelligent Systems
- Founder dan Koordinator **IlmuKomputer.Com**
- Peneliti LIPI (2004-2007)
- Founder dan CEO **PT Brainmatics Cipta Informatika**



Course Outline





3. Persiapan Data

3.1 Data Preprocessing

3.2 Data Cleaning

3.3 Data Reduction

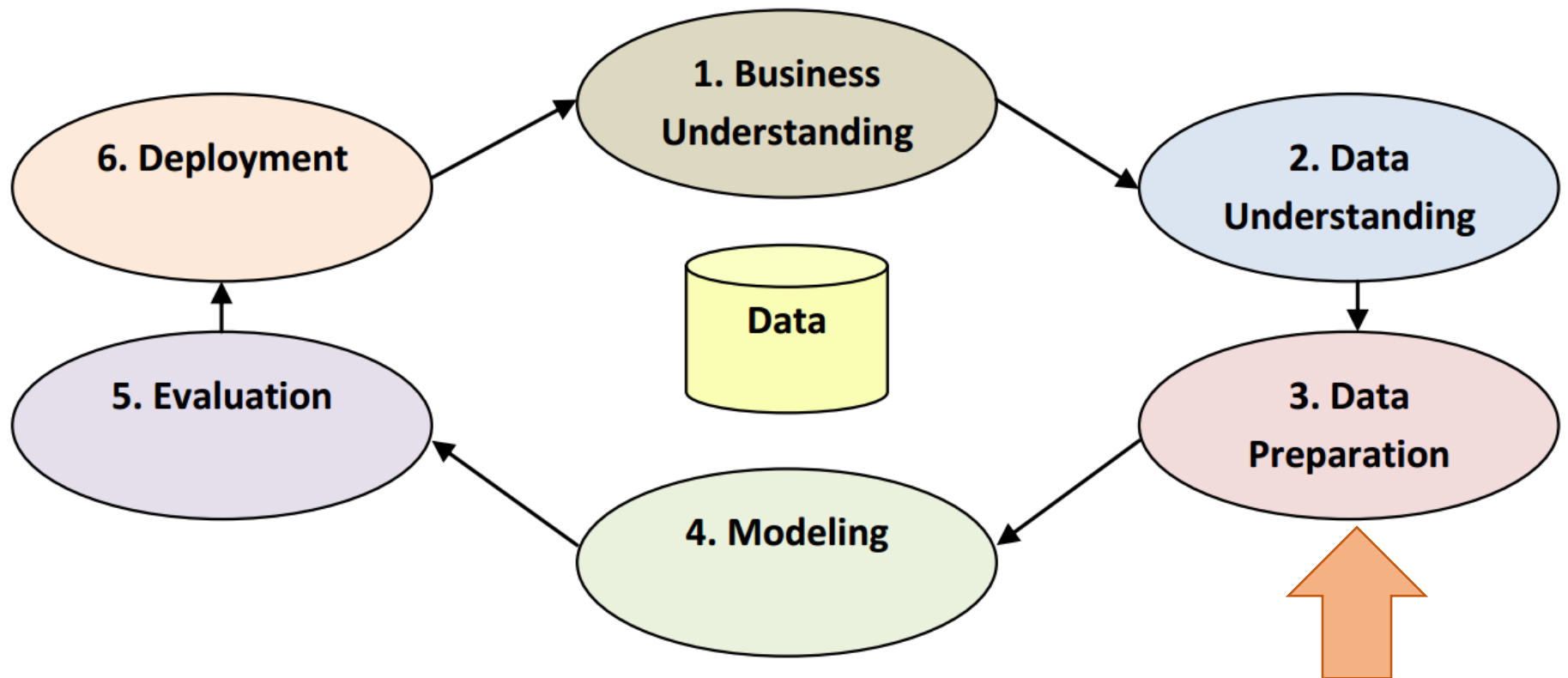
3.4 Data Transformation and Data Discretization

3.5 Data Integration



3.1 Data Preprocessing

CRISP-DM



Why Preprocess the Data?

Measures for **data quality**: A multidimensional view

- **Accuracy**: correct or wrong, accurate or not
- **Completeness**: not recorded, unavailable, ...
- **Consistency**: some modified but some not, ...
- **Timeliness**: timely update?
- **Believability**: how trustable the data are correct?
- **Interpretability**: how easily the data can be understood?

Major Tasks in Data Preprocessing

1. Data **cleaning**

- Fill in **missing** values
- Smooth **noisy** data
- Identify or **remove outliers**
- Resolve **inconsistencies**

2. Data **reduction**

- **Dimensionality** reduction
- **Numerosity** reduction
- Data **compression**

3. Data **transformation** and data **discretization**

- **Normalization**
- Concept hierarchy generation

4. Data **integration**

- Integration of **multiple databases** or files



3.2 Data Cleaning

Data Cleaning

Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error

- **Incomplete:** lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., Occupation=" " (**missing data**)
- **Noisy:** containing noise, errors, or outliers
 - e.g., Salary="-10" (**an error**)
- **Inconsistent:** containing discrepancies in codes or names
 - e.g., Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
- Discrepancy between **duplicate records**
 - Intentional (e.g., **disguised missing data**)
 - Jan. 1 as everyone's birthday?

Incomplete (Missing) Data

- Data is **not always available**
 - E.g., **many tuples have no recorded value** for several attributes, such as customer income in sales data
- **Missing data** may be due to
 - equipment **malfunction**
 - inconsistent with other recorded data and thus **deleted**
 - data not entered due to **misunderstanding**
 - certain data **may not be considered important** at the time of entry
 - not register history or **changes of the data**
- Missing data may **need to be inferred**

Contoh Missing Data

- Dataset: **MissingDataSet.csv**

ExampleSet (11 examples, 0 special attributes, 15 regular attributes)

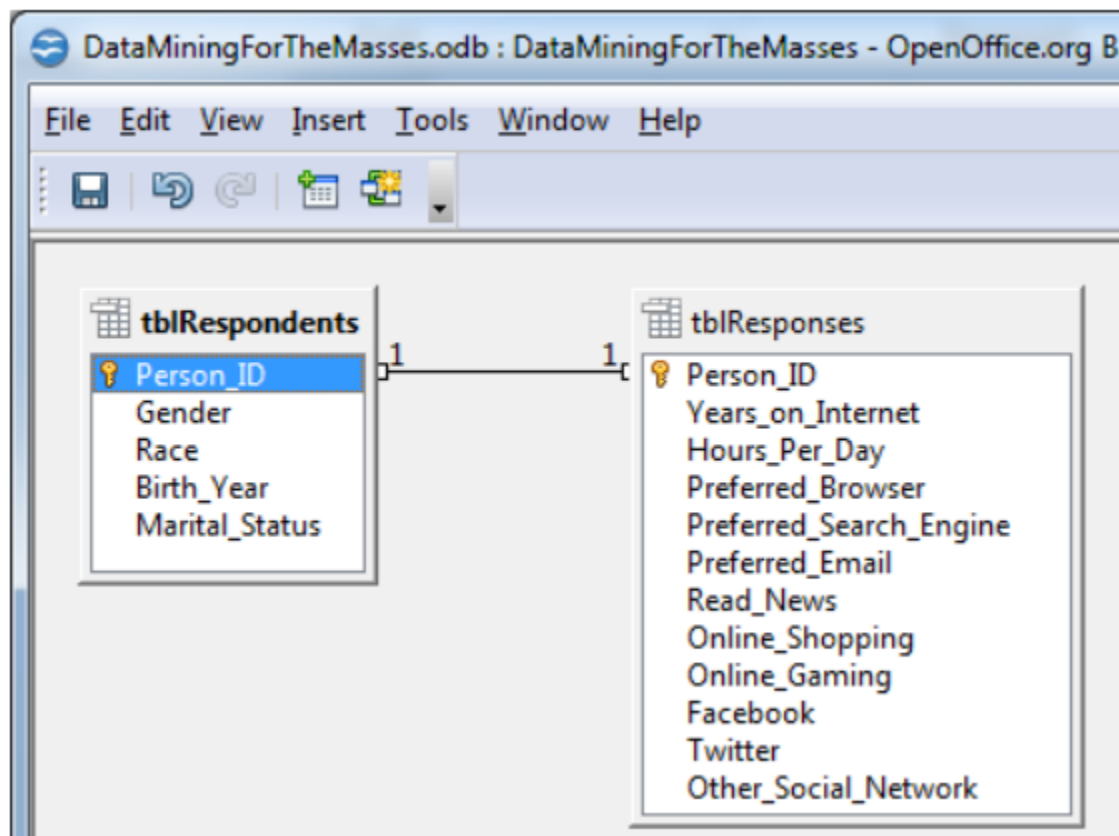
View Filter (11 / 11): all

Row No.	Gender	Race	Birth_Year	Marital_Stat...	Years_on_I...	Hours_Per...	Preferred_B...	Preferred_S...	Preferred_E...	Read_News	Online_Sho...	Online_Ga...	Facebook	Twitter	Other_Soci...
1	M	White	1972	M	8	1	Firefox	Google	Yahoo	Y	N	N	Y	N	?
2	M	Hispanic	1981	S	14	2	Chrome	Google	Hotmail	Y	N	N	Y	N	?
3	F	African Amer	1977	S	6	2	Firefox	Yahoo	Yahoo	Y	Y	?	Y	N	?
4	F	White	1961	D	8	6	Firefox	Google	Hotmail	N	Y	N	N	Y	?
5	M	White	1954	M	2	3	Internet Expl	Bing	Hotmail	Y	Y	N	Y	N	?
6	M	African Amer	1982	D	15	4	Internet Expl	Google	Yahoo	Y	N	Y	N	N	?
7	M	African Amer	1981	D	11	2	Firefox	Google	Yahoo	?	Y	?	Y	Y	LinkedIn
8	M	White	1977	S	3	3	Internet Expl	Yahoo	Yahoo	Y	?	?	Y	99	LinkedIn
9	F	African Amer	1969	M	6	2	Firefox	Google	Gmail	N	Y	N	N	N	?
10	M	White	1987	S	12	1	Safari	Yahoo	Yahoo	Y	?	Y	Y	N	MySpace
11	F	Hispanic	1959	D	12	5	Chrome	Google	Gmail	Y	N	N	Y	N	Google+

MissingDataSet.csv

- Jerry is the marketing manager for a **small Internet design and advertising firm**
- Jerry's boss asks him to develop a data set containing **information about Internet users**
- The company will use this data to determine **what kinds of people are using the Internet** and **how the firm may be able to market their services to this group of users**
- To accomplish his assignment, Jerry creates an online survey and places links to the survey on several popular Web sites
- Within two weeks, Jerry has collected enough data to begin analysis, but he finds that his data needs to be **denormalized**
- He also notes that some observations in the set are **missing values** or they appear to contain **invalid values**
- Jerry realizes that some additional work on the data needs to take place before analysis begins.

Relational Data



View of Data (Denormalized Data)

DataMiningForTheMasses.odt - View1 - OpenOffice.org Base: View Design

File Edit View Insert Tools Window Help

tblRespondents

Person_ID
Gender
Race
Birth_Year
Marital_Status

tblResponses

Person_ID
Years_on_Internet
Hours_Per_Day
Preferred_Browser
Preferred_Search_Engine
Preferred_Email
Read_News
Online_Shopping
Online_Gaming
Facebook
Twitter
Other_Social_Network

Field	Gender	Race	Birth_Year	Marital_Status	Years_on_Internet	Hours_Per_Day	Preferred_Browser	Preferred_Search_Engine	Preferred_Email	Read_News	Online_Shopping	Online_Gaming	Facebook	Twitter	Other_Social_Network
Alias															
Table	tblRespondents	tblRespondents	tblRespondents	tblRespondents	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses	tblResponses
Sort															
Visible	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Function															

vwInternetUser - DataMiningForTheMasses - OpenOffice.org Base: Table Data View

File Edit View Insert Tools Window Help

	Gender	Race	Birth_Year	Marital_Status	Years_on_Internet	Hours_Per_Day	Preferred_Browser	Preferred_Search_Engine	Preferred_Email	Read_News	Online_Shopping	Online_Gaming	Facebook	Twitter	Other_Social_Network
▶	M	White	1972	M	8	1	Firefox	Google	Yahoo	Y	N	N	Y	N	
	M	Hispanic	1981	S	14	2	Chrome	Google	Hotmail	Y	N	N	Y	N	
	F	African American	1977	S	6	2	Firefox	Yahoo	Yahoo	Y	Y		Y	N	
	F	White	1961	D	8	6	Firefox	Google	Hotmail	N	Y	N	N	Y	
	M	White	1954	M	2	3	Internet Explorer	Bing	Hotmail	Y	Y	N	Y	N	
	M	African American	1982	D	15	4	Internet Explorer	Google	Yahoo	Y	N	Y	N	N	
	M	African American	1981	D	11	2	Firefox	Google	Yahoo		Y		Y	Y	LinkedIn
	M	White	1977	S	3	3	Internet Explorer	Yahoo	Yahoo	Y			Y	99	LinkedIn
	F	African American	1969	M	6	2	Firefox	Google	Gmail	N	Y	N	N	N	
	M	White	1987	S	12	1	Safari	Yahoo	Yahoo	Y		Y	Y	N	MySpace
	F	Hispanic	1959	D	12	5	Chrome	Google	Gmail	Y	N	N	Y	N	Google+

Contoh Missing Data

- Dataset: **MissingDataSet.csv**

ExampleSet (11 examples, 0 special attributes, 15 regular attributes)

View Filter (11 / 11): all

Row No.	Gender	Race	Birth_Year	Marital_Stat...	Years_on_I...	Hours_Per...	Preferred_B...	Preferred_S...	Preferred_E...	Read_News	Online_Sho...	Online_Ga...	Facebook	Twitter	Other_Soci...
1	M	White	1972	M	8	1	Firefox	Google	Yahoo	Y	N	N	Y	N	?
2	M	Hispanic	1981	S	14	2	Chrome	Google	Hotmail	Y	N	N	Y	N	?
3	F	African Amer	1977	S	6	2	Firefox	Yahoo	Yahoo	Y	Y	?	Y	N	?
4	F	White	1961	D	8	6	Firefox	Google	Hotmail	N	Y	N	N	Y	?
5	M	White	1954	M	2	3	Internet Expl	Bing	Hotmail	Y	Y	N	Y	N	?
6	M	African Amer	1982	D	15	4	Internet Expl	Google	Yahoo	Y	N	Y	N	N	?
7	M	African Amer	1981	D	11	2	Firefox	Google	Yahoo	?	Y	?	Y	Y	LinkedIn
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10	M	White	1987	S	12	1	Safari	Yahoo	Yahoo	Y	?	Y	Y	N	MySpace
11	F	Hispanic	1959	D	12	5	Chrome	Google	Gmail	Y	N	N	Y	N	Google+

How to Handle Missing Data?

- **Ignore the tuple:**
 - Usually done when class **label is missing** (when doing classification)—not effective when the % of missing values per attribute varies considerably
- **Fill in the missing value manually:**
 - **Tedious + infeasible?**
- **Fill in it automatically** with
 - A **global constant**: e.g., “unknown”, a new class?!
 - The **attribute mean**
 - The **attribute mean for all samples belonging to the same class**: smarter
 - The **most probable value**: inference-based such as Bayesian formula or decision tree

Latihan

- Lakukan eksperimen mengikuti buku Matthew North, Data Mining for the Masses, 2012, **Chapter 3 Data Preparation**, pp. 30-46 (**Handling Missing Data**)
- Dataset: **MissingDataSet.csv**
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!

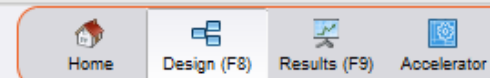
Missing Value Detection

Result Overview ExampleSet (Read CSV)

☐ Data View ☒ Meta Data View ☐ Plot View ☐ Advanced Charts ☐ Annotations

ExampleSet (11 examples, 0 special attributes, 15 regular attributes)

Role	Name	Type	Statistics	Range	Missings
regular	Gender	binominal	mode = M (7), least = F (4)	M (7), F (4)	0
regular	Race	polynomial	mode = White (5), least = Hispanic	White (5), Hispanic (2), African Ame	0
regular	Birth_Year	integer	avg = 1972.727 +/- 10.743	[1954.000 ; 1987.000]	0
regular	Marital_Status	polynomial	mode = S (4), least = M (3)	M (3), S (4), D (4)	0
regular	Years_on_Internet	integer	avg = 8.818 +/- 4.332	[2.000 ; 15.000]	0
regular	Hours_Per_Day	integer	avg = 2.818 +/- 1.601	[1.000 ; 6.000]	0
regular	Preferred_Browser	polynomial	mode = Firefox (5), least = Safari (1	Firefox (5), Chrome (2), Internet Exp	0
regular	Preferred_Search_Engine	polynomial	mode = Google (7), least = Bing (1	Google (7), Yahoo (3), Bing (1)	0
regular	Preferred_Email	polynomial	mode = Yahoo (6), least = Gmail (2	Yahoo (6), Hotmail (3), Gmail (2)	0
regular	Read_News	binominal	mode = Y (8), least = N (2)	Y (8), N (2)	1
regular	Online_Shopping	binominal	mode = Y (5), least = N (4)	N (4), Y (5)	2
regular	Online_Gaming	binominal	mode = N (6), least = Y (2)	N (6), Y (2)	3
regular	Facebook	binominal	mode = Y (8), least = N (3)	Y (8), N (3)	0
regular	Twitter	polynomial	mode = N (8), least = 99 (1)	N (8), Y (2), 99 (1)	0
regular	Other_Social_Network	polynomial	mode = LinkedIn (2), least = MySp	LinkedIn (2), MySpace (1), Google+	7



Operators

- repla
- Data Transformation (5)
 - Name and Role Modification
 - Rename by Replacing
 - Value Modification (2)
 - Nominal Value Modification
 - Replace (Dictionary)
 - Replace
 - Data Cleansing (2)
 - Replace Missing Values

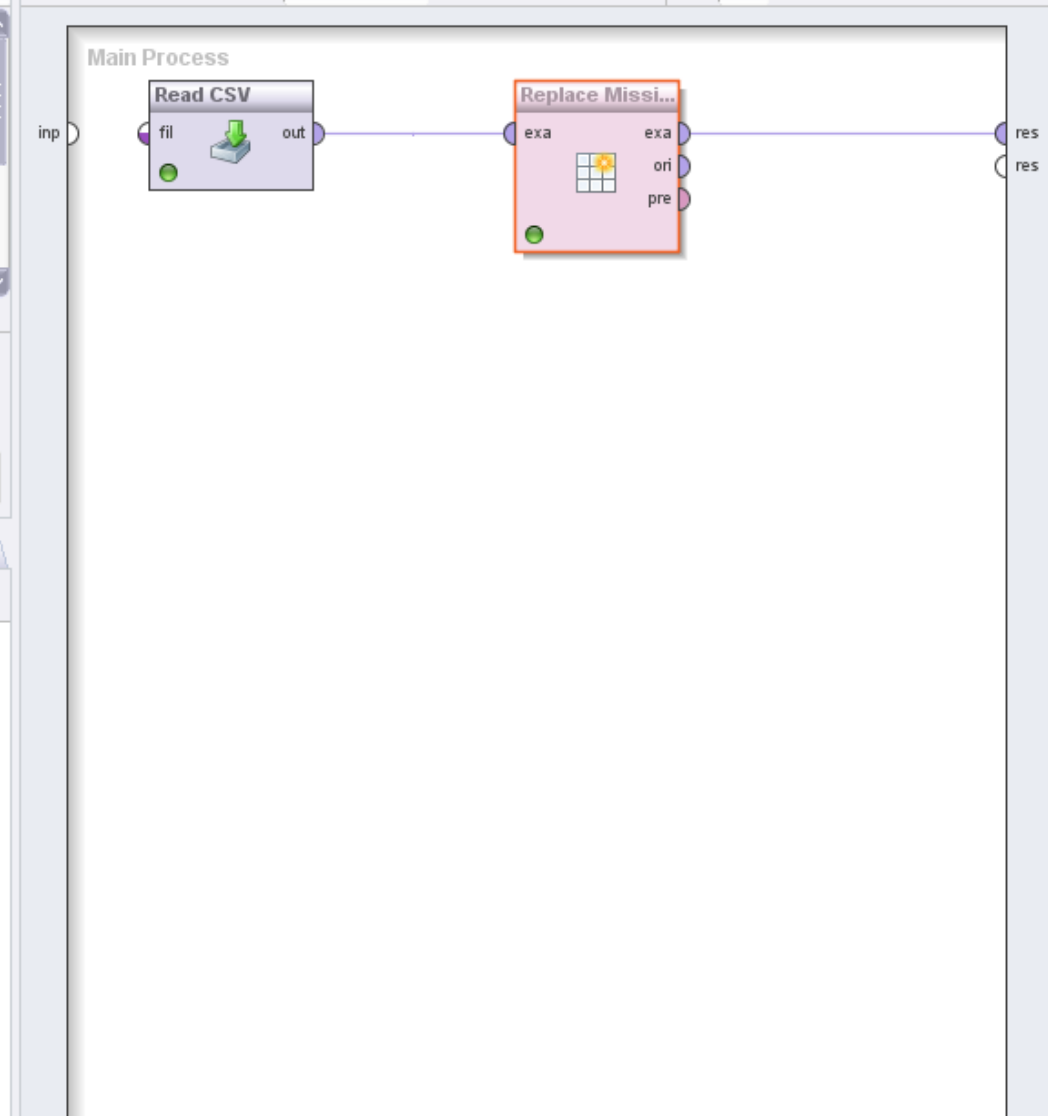
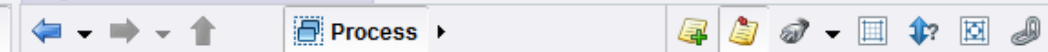
Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

☒ Activate Wisdom of Crowds

Repositories

- Samples
- DB
- Local Repository (RomiSatria)
- Cloud Repository (disconnected)

Process



Parameters



Replace Missing Values

☐ create view

attribute filter type: subset

attributes: Select Attribute...

☐ invert selection

☐ include special attributes

default: average

columns: Edit List (0)...

☒ Compatibility level: 6.5.002

Help

Replace Missing Values
(RapidMiner Studio Core)

Synopsis

This operator replaces missing values in examples of selected attributes by a

Noisy Data

- Noise: **random error** or variance in a measured variable
- **Incorrect attribute** values may be due to
 - Faulty data collection instruments
 - Data entry problems
 - Data transmission problems
 - Technology limitation
 - Inconsistency in naming convention
- **Other data problems** which require data cleaning
 - Duplicate records
 - Incomplete data
 - Inconsistent data

How to Handle Noisy Data?

- **Binning**

- First **sort data and partition** into (equal-frequency) bins
- Then one can **smooth by bin means**, smooth by bin median, smooth by bin boundaries, etc.

- **Regression**

- Smooth by **fitting the data into regression** functions

- **Clustering**

- Detect and **remove outliers**

- **Combined computer and human inspection**

- Detect suspicious values and **check by human** (e.g., deal with possible outliers)

Data Cleaning as a Process

- Data **discrepancy detection**
 - Use **metadata** (e.g., domain, range, dependency, distribution)
 - Check **field overloading**
 - Check **uniqueness rule**, consecutive rule and null rule
 - Use **commercial tools**
 - **Data scrubbing**: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - **Data auditing**: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)
- Data **migration** and integration
 - Data **migration tools**: allow transformations to be specified
 - **ETL (Extraction/Transformation/Loading) tools**: allow users to specify transformations through a graphical user interface
- Integration of the two processes
 - **Iterative and interactive** (e.g., Potter's Wheels)

Latihan

- Lakukan eksperimen mengikuti buku Matthew North, Data Mining for the Masses, 2012, **Chapter 3 Data Preparation**, pp. 50-52 (**Handling Inconsistence Data**)
- Dataset: **MissingDataSet.csv**
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!

<new process*> - RapidMiner Studio Community 7.0.001 @ RSW-BLUE

File Edit Process View Connections Cloud Settings Extensions

Views: Design Results

Questions?

Repository

+ Add Data

- DataPemilukPU (RomiSatria - v1, 2/24/16)
- HargaSaham (RomiSatria - v1, 2/24/16)
- HeatingOil (RomiSatria - v1, 2/24/16)
- HeatingOil-Scoring (RomiSatria - v1, 2/24/16)
- IMFCountry (RomiSatria - v1, 11/11/15)
- MusicGenre (RomiSatria - v1, 2/22/16)
- SportSkill (RomiSatria - v1, 2/24/16)
- SportSkill-Scoring (RomiSatria - v1, 2/24/16)
- Transaksi (RomiSatria - v1, 11/11/15)
- MissingValueData (RomiSatria - v1, 2/24/16)

Operators

repla

- Values (2)
 - Replace
 - Replace (Dictionary)
- Cleansing (2)
- Missing (2)
 - Replace Missing Values
 - Replace Infinite Values
- Extensions (3)
 - QuantX-1 Extension (1)

+ Get More Operators

Process

Process

inp

Retrieve MissingVal...

Replace Missing Val...

Replace

res

res

Parameters

Replace

attribute filter type: single

attribute: Twitter

☐ invert selection

☐ include special attributes

replace what: 9.*

replace by: N

[Change compatibility \(7.0.001\)](#)

Help

Replace

RapidMiner Studio Core

Synopsis

This operator replaces parts of the values of selected nominal attributes matching a specified regular expression by a specified replacement.

Latihan

- Lakukan eksperimen mengikuti buku Matthew North, Data Mining for the Masses, 2012, **Chapter 8 Estimation**, pp. 127-140 (**Estimation**)
- Dataset: **HeatingOil.csv**
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!

) special attributes, 6 regular attributes):

Name	Type	Statistics	Range	
Insulation	integer	avg = 6.214 +/- 2.768	[2.000 ; 10.000]	0
Temperature	integer	avg = 65.079 +/- 16.932	[38.000 ; 90.000]	0
Heating_Oil	integer	avg = 197.394 +/- 56.248	[114.000 ; 301.000]	0
Num_Occupants	integer	avg = 3.113 +/- 1.691	[1.000 ; 10.000]	0
Avg_Age	real	avg = 42.706 +/- 15.051	[15.100 ; 72.200]	0
Home_Size	integer	avg = 4.649 +/- 2.321	[1.000 ; 8.000]	0

Name	Type	Statistics	Range	
Insulation	integer	avg = 5.989 +/- 2.576	[2.000 ; 10.000]	0
Temperature	integer	avg = 63.962 +/- 15.313	[38.000 ; 90.000]	0
Num_Occupants	integer	avg = 5.489 +/- 2.875	[1.000 ; 10.000]	0
Avg_Age	real	avg = 44.040 +/- 16.737	[15.000 ; 73.000]	0
Home_Size	integer	avg = 4.495 +/- 2.291	[1.000 ; 8.000]	0



3.3 Data Reduction

Data Reduction Strategies

- Data Reduction

- Obtain a **reduced representation of the data set** that is much smaller in volume but yet produces the same analytical results

- Why Data Reduction?

- A database/data warehouse may store **terabytes of data**
- Complex data analysis **take a very long time to run** on the complete dataset

- Data Reduction Strategies

1. Dimensionality reduction

1. Feature Extraction
2. Feature Selection

2. Numerosity reduction (Data Reduction)

- Regression and Log-Linear Models
- Histograms, clustering, sampling

1. Dimensionality Reduction

- Curse of **dimensionality**
 - When dimensionality increases, **data becomes increasingly sparse**
 - Density and distance between points, which is critical to clustering, outlier analysis, **becomes less meaningful**
 - The possible combinations of subspaces will grow exponentially
- Dimensionality **reduction**
 - Avoid the curse of dimensionality
 - Help **eliminate irrelevant features** and reduce noise
 - **Reduce time and space** required in data mining
 - Allow easier visualization
- Dimensionality **reduction techniques**
 1. **Feature Extraction**: Wavelet transforms, Principal Component Analysis (PCA)
 2. **Feature Selection**: Filter, Wrapper, Embedded

Principal Component Analysis (Steps)

- Given N data vectors from n -dimensions, **find $k \leq n$ orthogonal vectors (*principal components*)** that can be best used to represent data
 1. **Normalize input data**: Each attribute falls within the same range
 2. **Compute k orthonormal (unit) vectors**, i.e., *principal components*
 3. Each input data (vector) is a linear combination of the k principal component vectors
 4. The principal components are **sorted in order of decreasing “significance”** or strength
 5. Since the components are sorted, the size of the data can be reduced by **eliminating the *weak components***, i.e., those with low variance
- Works for **numeric data only**

Latihan

- Lakukan eksperimen mengikuti buku Markus Hofmann (Rapid Miner - Data Mining Use Case) **Chapter 4 (k-Nearest Neighbor Classification II)** pp. 45-51
- Dataset: [glass.data](#)
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!
- Bandingkan **akurasi** dari k-NN dan PCA+k-NN

<new process> - RapidMiner Studio Community 7.0.001 @ RSW-BLUE

File Edit Process View Connections Cloud Settings Extensions

Views: Design Results

Questions?

Repository

+ Add Data

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- Transaksi (RomiSatria - v1, 2/25/2019)
- MissingValueData (RomiSatria - v1, 2/25/2019)
- Glass (RomiSatria - v1, 2/25/2019)

Operators

Data Editor

Row No.	Id (integer) id
1	1
2	2
3	3
4	4
5	5
6	6

Repository Location: //Local R...

Process

Process

Retrieve Glass

PCA

Split Data

k-NN

Apply Model

Performance

Parameters

Split Data

partitions Edit ...

sampling ... strati...

use local random s...

Edit Parameter List: partitions

Edit Parameter List: **partitions**
The partitions that should be created.

ratio

0.7

0.3



Views:

Design

Results

Questions? ▾

Result History × ExampleSet (Split Data) × PerformanceVector (Performance) ×



Data



Statistics



Charts

Advanced
Charts

Annotations

ExampleSet (65 examples, 9 special attributes, 5 regular attributes)

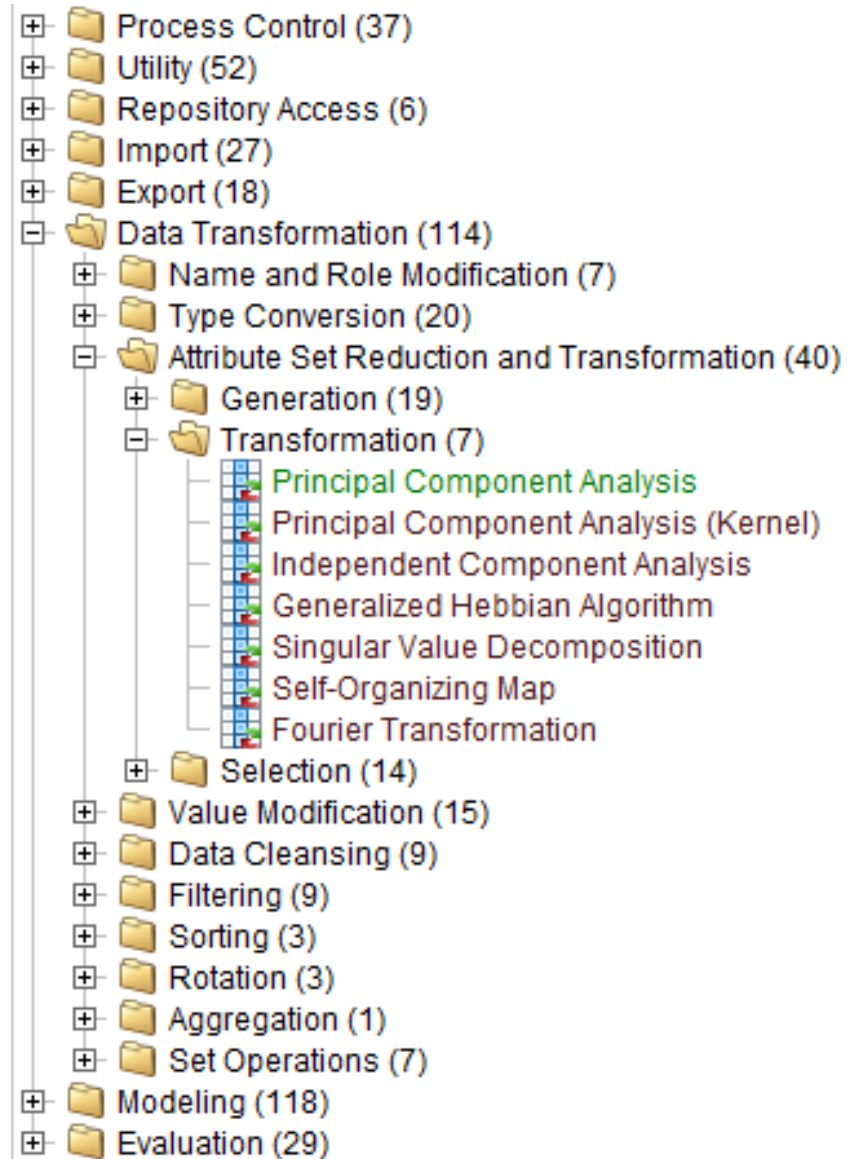
Filter (65 / 65 examples):

all ▾

confidence(2)	confidence(3)	confidence(5)	confidence(6)	confidence(7)	pc_1	pc_2	pc_3	pc_4	pc_5
1	0	0	0	0	-1.437	0.344	0.278	0.294	0.194
1	0	0	0	0	-1.427	0.346	-0.139	0.322	-0.024
0	0	0	0	0	-1.312	-0.018	-0.358	0.279	0.071
0	0	0	0	0	-1.049	-0.324	-0.761	0.202	-0.050
0	0	0	0	0	-0.781	-0.585	0.909	0.353	0.038
1	0	0	0	0	-0.949	-0.441	-0.411	-0.111	-0.196
0	0	0	0	0	-0.978	-0.205	-0.098	0.251	0.127
0	0	0	0	0	-0.956	-0.387	-0.487	0.090	0.044
0	0	0	0	0	-0.926	-0.472	-0.862	0.023	-0.215
0	0	0	0	0	-0.937	-0.310	-0.327	0.052	0.023
0	0	0	0	0	-0.860	-0.596	-0.740	0.033	-0.123
0	0	0	0	0	-0.807	-0.515	-0.618	-0.045	-0.090
0	0	0	0	0	-0.413	-1.176	1.448	0.465	0.411
0	1	0	0	0	-0.561	-0.619	0.617	-0.283	0.079
0	0	0	0	0	-0.618	-0.469	-0.136	0.063	0.096
0	0	0	0	0	-0.076	-1.698	0.648	0.163	-0.021
0	1	0	0	0	-0.537	-0.459	0.560	-0.190	0.191
0	0	0	0	0	-0.012	-1.618	1.167	0.301	0.227
0	0	0	0	0	-0.075	-0.200	-0.354	0.185	0.123

Latihan

- Ganti PCA dengan **metode dimension reduction** yang lain
- Cek di RapidMiner, **operator apa saja yang bisa digunakan** untuk mengurangi dimensi dari dataset



Feature/Attribute Selection

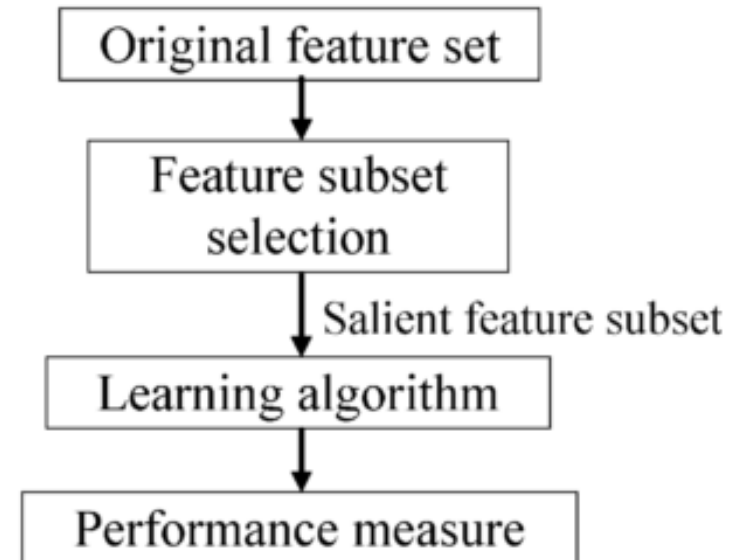
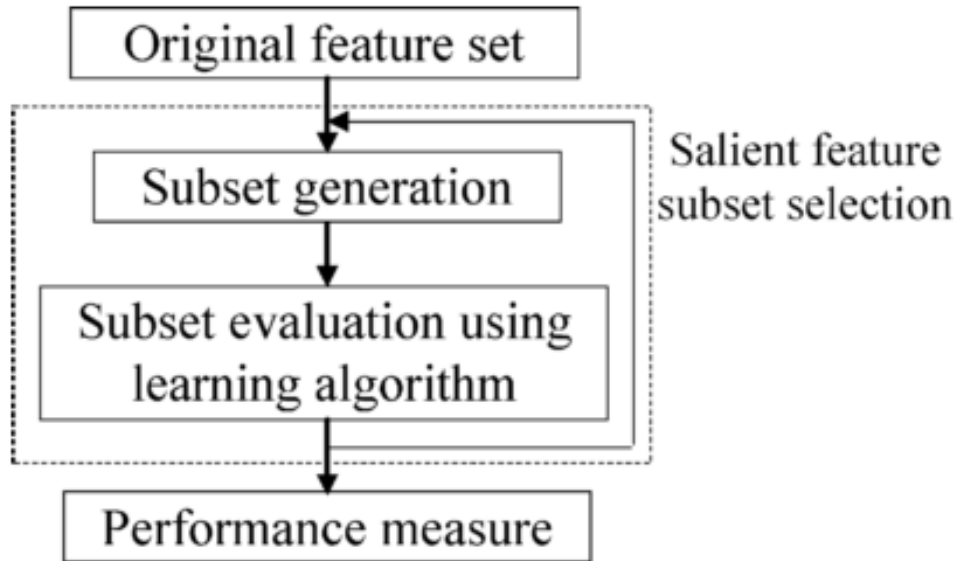
- Another way to reduce dimensionality of data
- **Redundant** attributes
 - Duplicate much or all of the information contained in one or more other attributes
 - E.g., purchase price of a product and the amount of sales tax paid
- **Irrelevant** attributes
 - Contain no information that is useful for the data mining task at hand
 - E.g., students' ID is often irrelevant to the task of predicting students' GPA

Feature Selection Approach

A number of proposed approaches for feature selection can broadly be categorized into the following three classifications: **wrapper**, **filter**, and **hybrid** (Liu & Tu, 2004)

1. In the **filter approach**, statistical analysis of the feature set is required, **without utilizing any learning model** (Dash & Liu, 1997)
2. In the **wrapper approach**, a predetermined learning model is assumed, wherein **features are selected that justify the learning performance** of the particular learning model (Guyon & Elisseeff, 2003)
3. The **hybrid approach** attempts to utilize the complementary strengths of the wrapper and filter approaches (Huang, Cai, & Xu, 2007)

Wrapper Approach vs Filter Approach



Feature Selection Approach

1. Filter Approach:

- information gain
- chi square
- log likelihood ratio



2. Wrapper Approach:

- forward selection
- backward elimination
- randomized hill climbing

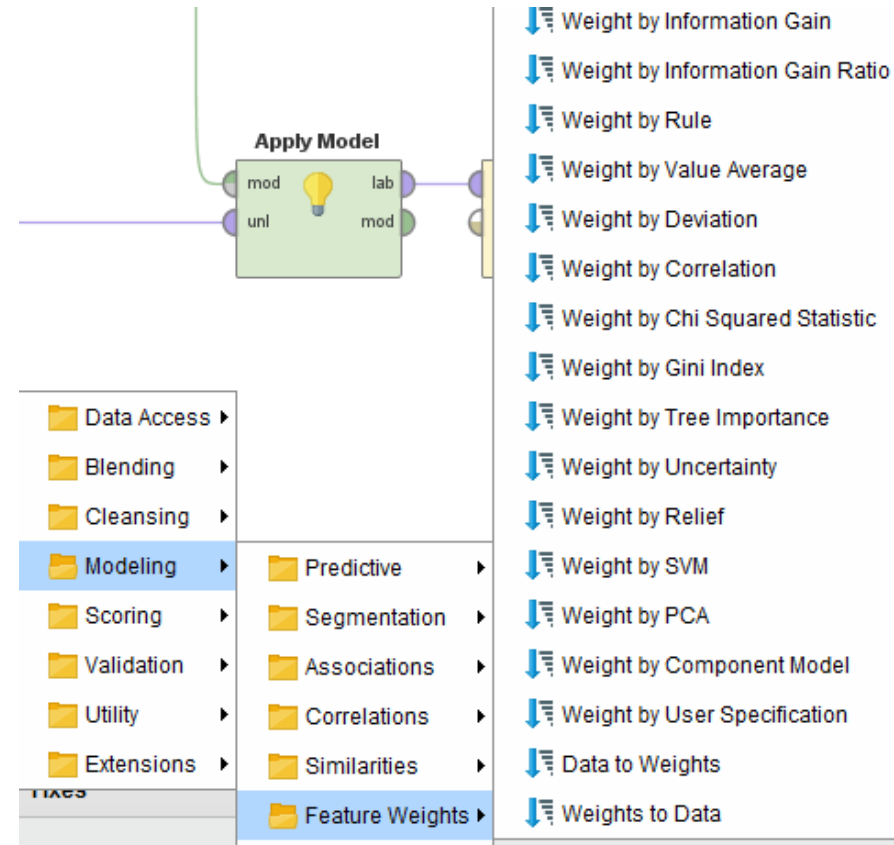


3. Embedded Approach:

- decision tree
- weighted naïve bayes

Latihan

- Lakukan eksperimen mengikuti buku Markus Hofmann (Rapid Miner - Data Mining Use Case) Chapter 4 (k-Nearest Neighbor Classification II)
- Ganti PCA dengan metode feature selection (filter), misalnya:
 - Information Gain
 - Chi Squared
 - etc
- Cek di RapidMiner, operator apa saja yang bisa digunakan untuk mengurangi atau membobot attribute dari dataset!



< new process* - RapidMiner Studio Community 7.0.001 @ RSW-BLUE

File Edit Process View Connections Cloud Settings Extensions

Views: Design Results Questions?

Repository

Add ...

- SportSkill-Scoring
- Transaksi (RomiSatria)
- MissingValueData
- Glass (RomiSatria)
- hofmann (RomiSatria)
- kotu (RomiSatria)
- processes (RomiSatria)
- Federalis Paper (RomiSatria)
- HeatingOil-Modeling
- HeatingOil-Scoring (RomiSatria)

Operators

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- Extract Perform
- Combine Perfo
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+ Get More Operators

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encoding SYSTEM

[Hide advanced parameters](#)

[Change compatibility \(7.0.001\)](#)

Help

Process

Synopsis

The root operator which is the outer most operator of every process.

Latihan

- Lakukan eksperimen mengikuti buku Markus Hofmann (Rapid Miner - Data Mining Use Case) **Chapter 4 (k-Nearest Neighbor Classification II)**
- Ganti PCA dengan **metode feature selection (wrapper)**, misalnya:
 - Backward Elimination
 - Forward Selection
 - etc
- Ganti metode validasi dengan 10-Fold X Validation
- Bandingkan **akurasi** dari k-NN dan BE+k-NN or FS+k-NN

<new process*> - RapidMiner Studio Community 7.0.001 @ RSW-BLUE

File Edit Process View Connections Cloud Settings Extensions

Views: Design Results

Questions?

Repository

+ Add Data

- SportSkill-Scoring (RomiSatria - v1, 2/24/...
- Transaksi (RomiSatria - v1, 11/11/15 2:10...
- MissingValueData (RomiSatria - v1, 2/25/...
- Glass (RomiSatria - v1, 2/25/18 10:38 AM)
- hofmann (RomiSatria)
- kotu (RomiSatria)

Operators

Search for Operators

- Parameters (5)
- Feature Selection (6)
 - Forward Selection
 - Backward Elimination
 - Optimize Selection
 - Optimize Selection (Brute)
 - Optimize Selection (Weight)
 - Optimize Selection (Evolut
- Feature Generation (5)
- Feature Weighting (4)
- Scoring (10)
- Validation (30)
- Utility (87)

+ Get More Operators

Process

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Parameters

Process

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logfile:

resultfile:

random seed: 2001

send mail: never

encoding: SYSTEM

[Hide advanced parameters](#)

[Change compatibility \(7.0.001\)](#)

Help

Process

Synopsis

The root operator which is the outer most operator of every process.

Latihan: Prediksi Kelulusan Mahasiswa

1. Lakukan **training** pada data mahasiswa ([datakelulusanmahasiswa.xls](#)) dengan menggunakan Decision Tree (DT)
2. Lakukan feature selection dengan **Forward Selection** untuk algoritma DT (DT+FS)
3. Lakukan feature selection dengan **Backward Elimination** untuk algoritma DT (DT+BE)
4. Lakukan **pengujian** dengan menggunakan 10-fold X Validation
5. Uji beda dengan **t-Test** untuk mendapatkan model terbaik (DT vs DT+FS vs DT+BE)

	DT	DT+FS	DT+BE
Accuracy	91.29	92.63	91.81
AUC	0.893	0.919	0.906

Latihan: Prediksi Kelulusan Mahasiswa

1. Lakukan **training** pada data mahasiswa ([datakelulusanmahasiswa.xls](#)) dengan menggunakan DT, NB, K-NN
2. Lakukan dimension reduction dengan **Forward Selection** untuk ketiga algoritma di atas
3. Lakukan **pengujian** dengan menggunakan 10-fold X Validation
4. Uji beda dengan **t-Test** untuk mendapatkan model terbaik

	DT	NB	K-NN	DT+FS	NB+FS	K-NN+FS
Accuracy						
AUC						

Latihan

- Lakukan **training** pada data eReader Adoption (eReader-Training.csv) dengan menggunakan DT dengan 3 alternative **criterion** (Gain Ratio, Information Gain dan Gini Index)
- Lakukan feature selection dengan **Forward Selection** untuk ketiga algoritma di atas
- Lakukan pengujian dengan menggunakan 10-fold X Validation
- Dari model terbaik, tentukan **faktor (atribut) apa saja yang berpengaruh** pada tingkat adopsi eReader

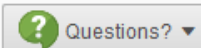
	DTGR	DTIG	DTGI	DTGR+FS	DTIG+FS	DTGI+FS
Accuracy	58.39	51.01	31.01	61.41	56.73	31.01



Views:

Design

Results



PerformanceVector (Performance (3)) ×

PerformanceVector (Performance (2)) ×

PerformanceVector (Performance) ×

PerformanceVector (Performance (6)) ×

PerformanceVector (Performance (5)) ×

PerformanceVector (Performance (4)) ×

Result History ×

AttributeWeights (Forward Selection) ×

AttributeWeights (Forward Selection) ×



Data



Charts



Annotations

attribute	weight ↓
Age	1
Website_Activity	1
Browsed_Electronics_12Mo	1
Bought_Digital_Media_18Mo	1
Bought_Digital_Books	1
Payment_Method	1
Gender	0
Marital_Status	0
Bought_Electronics_12Mo	0

2. Numerosity Reduction

Reduce data volume by choosing alternative, **smaller forms of data representation**

1. **Parametric methods** (e.g., regression)

- Assume the **data fits some model**, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
- Ex.: **Log-linear models**—obtain value at a point in m -D space as the product on appropriate marginal subspaces

2. **Non-parametric methods**

- **Do not assume models**
- Major families: **histograms, clustering**, sampling, ...

Parametric Data Reduction: Regression and Log-Linear Models

- **Linear regression**

- Data modeled to fit a straight line
- Often uses the least-square method to fit the line

- **Multiple regression**

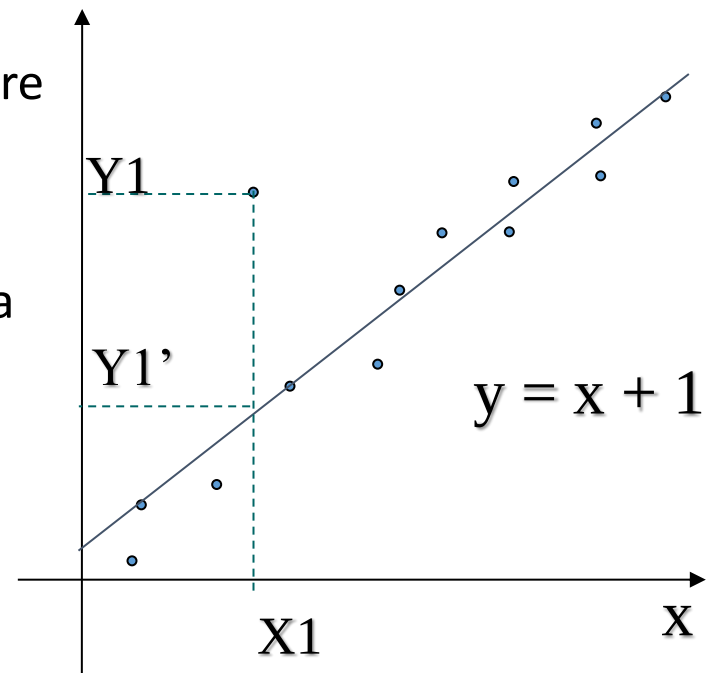
- Allows a response variable Y to be modeled as a linear function of multidimensional feature vector

- **Log-linear model**

- Approximates discrete multidimensional probability distributions

Regression Analysis

- **Regression analysis**: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a **dependent variable** (also called **response variable** or *measurement*) and of one or more *independent variables* (aka. **explanatory variables** or **predictors**)
- The parameters are estimated so as to give a "**best fit**" of the data
- Most commonly the best fit is evaluated by using the **least squares method**, but other criteria have also been used
- Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

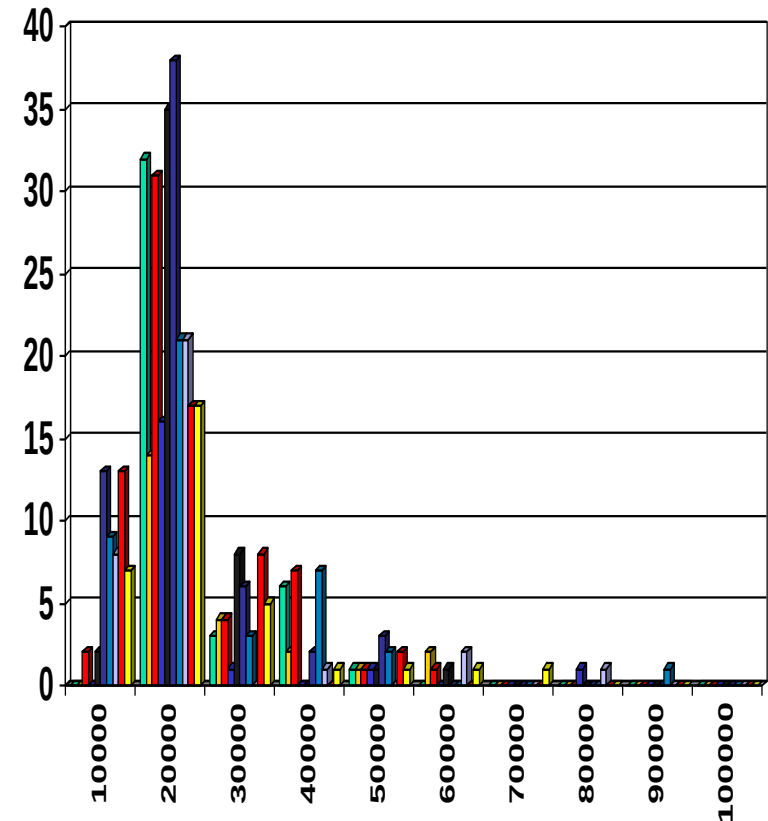


Regress Analysis and Log-Linear Models

- **Linear regression:** $Y = wX + b$
 - Two regression coefficients, w and b , specify the line and are to be estimated by using the data at hand
 - Using the least squares criterion to the known values of $Y_1, Y_2, \dots, X_1, X_2, \dots$
- **Multiple regression:** $Y = b_0 + b_1 X_1 + b_2 X_2$
 - Many nonlinear functions can be transformed into the above
- **Log-linear models:**
 - Approximate discrete multidimensional probability distributions
 - Estimate the probability of each point (tuple) in a multi-dimensional space for a set of discretized attributes, based on a smaller subset of dimensional combinations
 - Useful for dimensionality reduction and data smoothing

Histogram Analysis

- Divide data into buckets and store average (sum) for each bucket
- Partitioning **rules**:
 - **Equal-width**: equal bucket range
 - **Equal-frequency** (or equal-depth)



Clustering

- Partition data set into **clusters based on similarity**, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is clustered but not if data is “smeared”
- Can have hierarchical clustering and be stored in multi-dimensional index tree structures
- There are **many choices of clustering definitions** and clustering algorithms

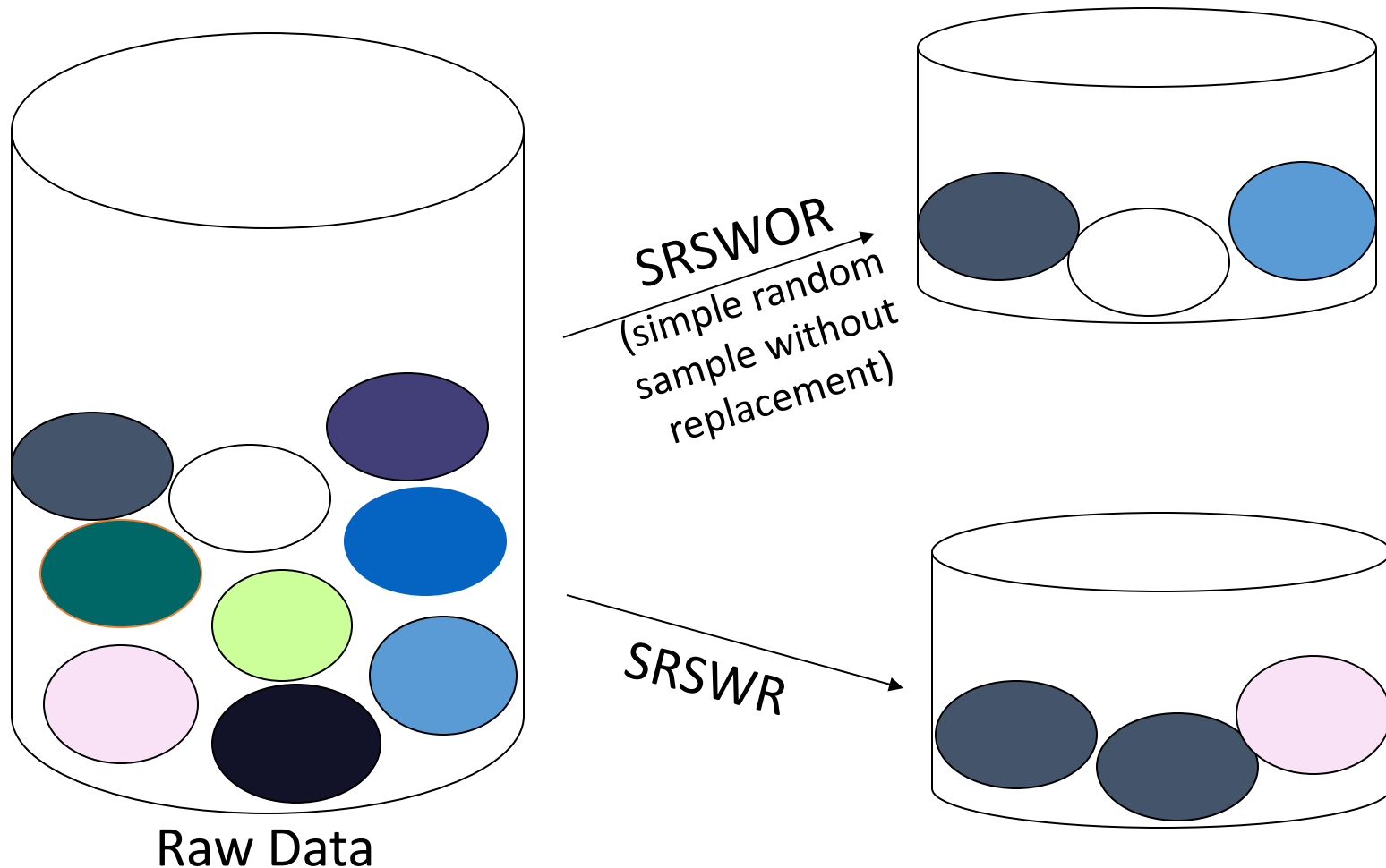
Sampling

- **Sampling**: obtaining a small sample s to represent the whole data set N
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- **Key principle**: Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
 - Develop adaptive sampling methods, e.g., stratified sampling
- **Note**: Sampling may not reduce database I/Os (page at a time)

Types of Sampling

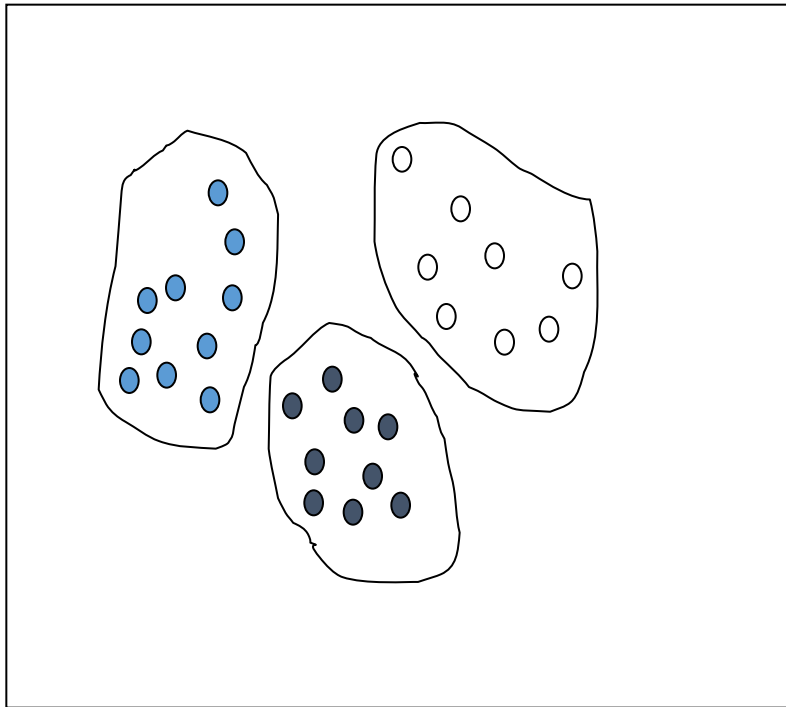
- **Simple random sampling**
 - There is an equal probability of selecting any particular item
- **Sampling without replacement**
 - Once an object is selected, it is removed from the population
- **Sampling with replacement**
 - A selected object is not removed from the population
- **Stratified sampling**
 - Partition the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)
 - Used in conjunction with skewed data

Sampling: With or without Replacement

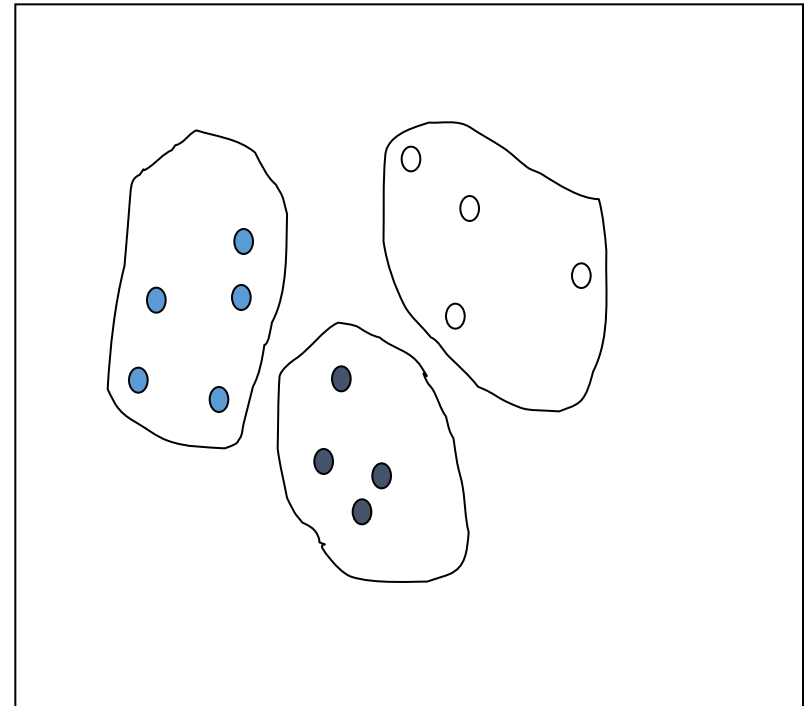


Sampling: Cluster or Stratified Sampling

Raw Data



Cluster/Stratified Sample



Stratified Sampling

- **Stratification** is the process of **dividing members of the population into homogeneous subgroups** before sampling
- Suppose that in a company there are the following **staff**:
 - Male, full-time: 90
 - Male, part-time: 18
 - Female, full-time: 9
 - Female, part-time: 63
 - Total: 180
- We are asked to take **a sample of 40 staff, stratified** according to the above categories
- An easy way to calculate the percentage is to multiply each group size by the sample size and divide by the total population:
 - Male, full-time = $90 \times (40 \div 180) = 20$
 - Male, part-time = $18 \times (40 \div 180) = 4$
 - Female, full-time = $9 \times (40 \div 180) = 2$
 - Female, part-time = $63 \times (40 \div 180) = 14$

Latihan

- Lakukan eksperimen mengikuti buku Matthew North, Data Mining for the Masses, 2012, **Chapter 7 Discriminant Analysis**, pp. 105-125
- Datasets: **SportSkill-Training.csv** dan **SportSkill-Scoring.csv**
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!

Latihan

- Lakukan eksperimen mengikuti buku Matthew North, Data Mining for the Masses, 2012, **Chapter 3 Data Preparation**, pp. 46-50 (**Data Reduction**)
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut



3.4 Data Transformation and Data Discretization

Data Transformation

- A function that **maps the entire set of values of a given attribute** to a new set of replacement values
 - Each old value **can be identified with one of the new values**
- **Methods:**
 - **Smoothing:** Remove noise from data
 - **Attribute/feature construction**
 - New attributes constructed from the given ones
 - **Aggregation:** Summarization, data cube construction
 - **Normalization:** Scaled to fall within a smaller, specified range
 - min-max normalization
 - z-score normalization
 - normalization by decimal scaling
 - **Discretization:** Concept hierarchy climbing

Normalization

- **Min-max normalization**: to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Ex. Let income range \$12,000 to \$98,000 normalized to $[0.0, 1.0]$.

Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- **Z-score normalization** (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

- Ex. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- **Normalization by decimal scaling**

$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

Discretization

- Three **types of attributes**
 - **Nominal** —values from an unordered set, e.g., color, profession
 - **Ordinal** —values from an ordered set, e.g., military or academic rank
 - **Numeric** —real numbers, e.g., integer or real numbers
- **Discretization**: Divide the range of a **continuous attribute into intervals**
 - Interval labels can then be used to replace actual data values
 - **Reduce data size** by discretization
 - Supervised vs. unsupervised
 - Split (top-down) vs. merge (bottom-up)
 - Discretization can be performed recursively on an attribute
 - Prepare for further analysis, e.g., classification

Data Discretization Methods

Typical **methods**: All the methods can be applied recursively

- **Binning**: Top-down split, unsupervised
- **Histogram analysis**: Top-down split, unsupervised
- **Clustering analysis**: Unsupervised, top-down split or bottom-up merge
- **Decision-tree analysis**: Supervised, top-down split
- **Correlation (e.g., χ^2) analysis**: Unsupervised, bottom-up merge

Simple Discretization: Binning

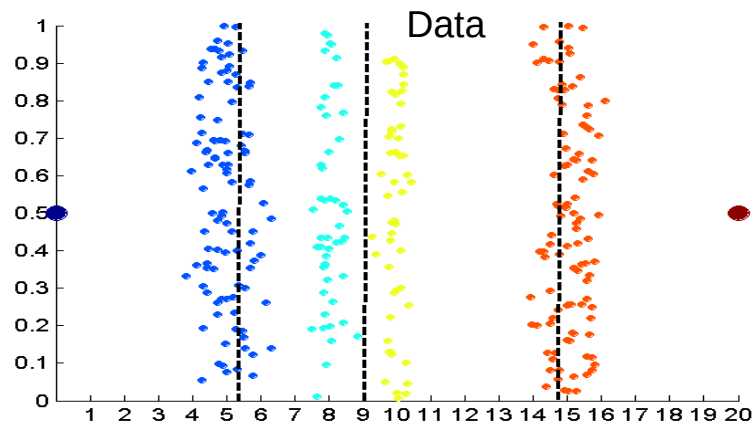
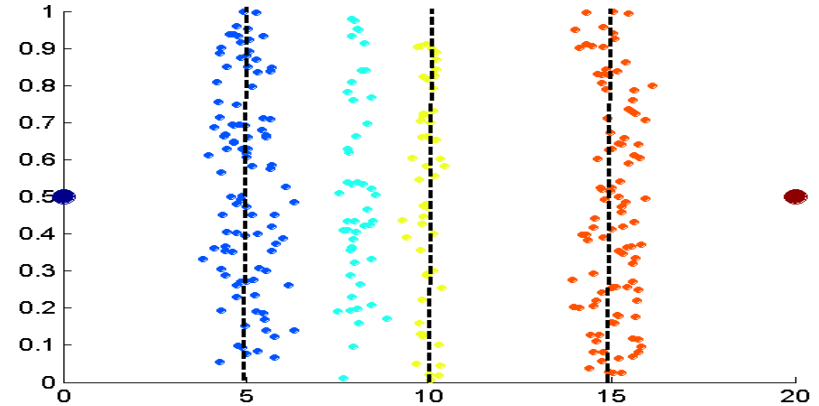
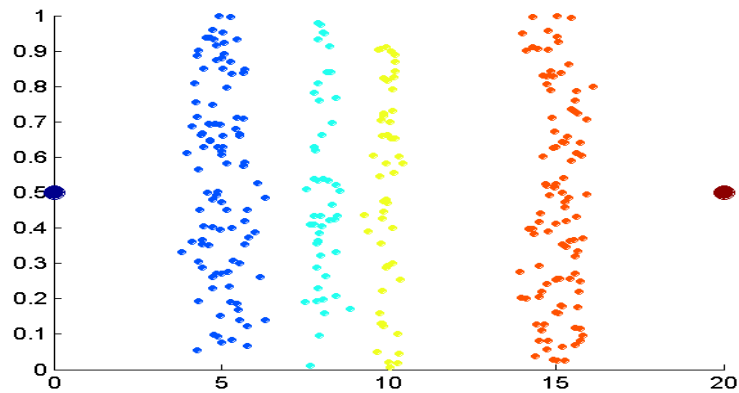
- **Equal-width** (distance) partitioning
 - Divides the range into N intervals of equal size: uniform grid
 - if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
 - The most straightforward, but outliers may dominate presentation
 - Skewed data is not handled well
- **Equal-depth** (frequency) partitioning
 - Divides the range into N intervals, each containing approximately same number of samples
 - Good data scaling
 - Managing categorical attributes can be tricky

Binning Methods for Data Smoothing

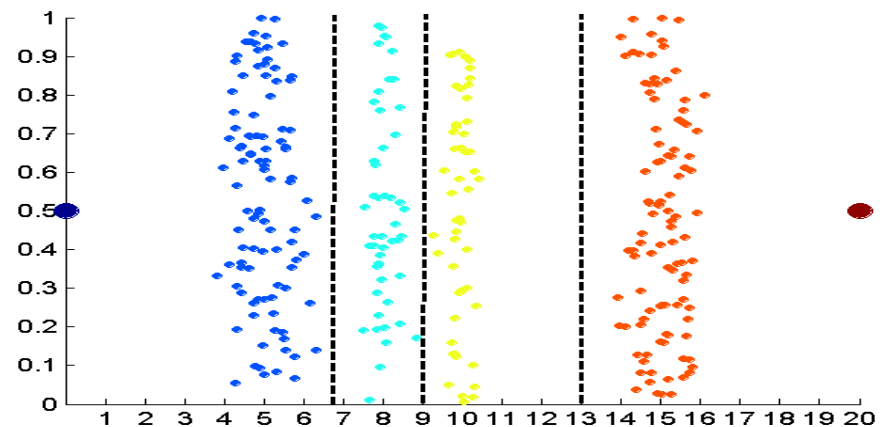
Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

- Partition into equal-frequency (**equi-depth**) bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
- Smoothing by **bin means**:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
- Smoothing by **bin boundaries**:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34

Discretization Without Using Class Labels (Binning vs. Clustering)



Equal frequency (binning)



K-means clustering leads to better results

Discretization by Classification & Correlation Analysis

- **Classification** (e.g., decision tree analysis)
 - Supervised: Given class labels, e.g., cancerous vs. benign
 - Using *entropy* to determine split point (discretization point)
 - Top-down, recursive split
- **Correlation analysis** (e.g., Chi-merge: χ^2 -based discretization)
 - Supervised: use class information
 - Bottom-up merge: find the best neighboring intervals (those having similar distributions of classes, i.e., low χ^2 values) to merge
 - Merge performed recursively, until a predefined stopping condition

Latihan

- Lakukan eksperimen mengikuti buku Markus Hofmann (Rapid Miner - Data Mining Use Case) **Chapter 5 (Naïve Bayes Classification I)**
- Dataset: **crx.data**
- Analisis **metode preprocessing** apa saja yang digunakan dan mengapa perlu dilakukan pada dataset tersebut!
- Bandingkan akurasi model apabila tidak menggunakan filter dan **diskretisasi**
- Bandingkan pula apabila digunakan feature selection (wrapper) dengan **Backward Elimination**

 Add Data

© 2006 The Authors

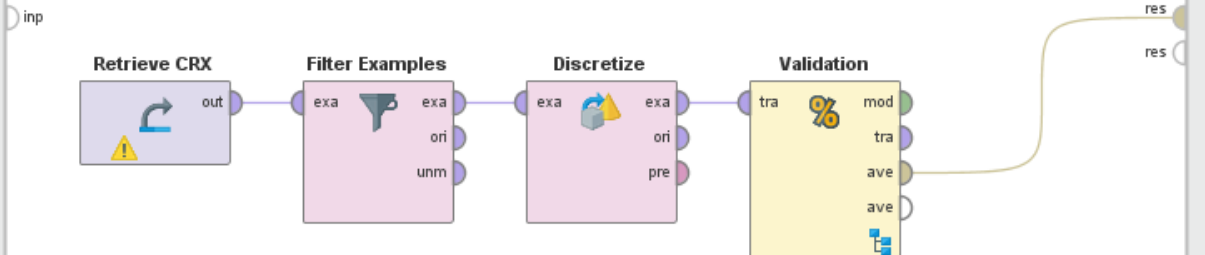
▼ Extensions (4)

▼ Series (3)

+ Get More Operators

- Process ▶

Process



A cross-validation
evaluating a
decision tree
model.

 Process Hide advanced parameters

✓ [Change compatibility \(7.0.001\)](#)

 Process

The root operator which is the outer most operator of every process.

Each process must contain exactly one operator of this class, and it must be the root operator of the process. This operator provides a set of

Hasil

	NB	NB+ Filter	NB+ Discretization	NB+ Filter+ Discretization	NB+ Filter+ Discretization + Backward Elimination
Accuracy				85.79	86.26
AUC					



3.5 Data Integration

Data Integration

- **Data integration:**
 - Combines data from **multiple sources** into a **coherent** store
- **Schema Integration:** e.g., $A.cust-id \equiv B.cust-\#$
 - Integrate **metadata** from different sources
- **Entity Identification Problem:**
 - Identify real world **entities** from **multiple data sources**, e.g., Bill Clinton = William Clinton
- **Detecting and Resolving Data Value Conflicts**
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales, e.g., metric vs. British units

Handling Redundancy in Data Integration

- Redundant data occur often **when integration of multiple databases**
 - **Object identification**: The same attribute or object may have different names in different databases
 - **Derivable data**: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- **Redundant attributes** may be able to be detected by **correlation analysis** and covariance analysis
- Careful integration of the data from multiple sources may help **reduce/avoid redundancies** and inconsistencies and improve mining speed and quality

Correlation Analysis (Nominal Data)

- χ^2 (chi-square) test

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

- The larger the χ^2 value, the **more likely the variables are related**
- The cells that contribute the most to the χ^2 value are those whose **actual count is very different from the expected count**
- Correlation does **not imply causality**
 - # of hospitals and # of car-theft in a city are correlated
 - Both are causally linked to the third variable: population

Chi-Square Calculation: An Example

	Play chess	Not play chess	Sum (row)
Like science fiction	250(90)	200(360)	450
Not like science fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

- χ^2 (chi-square) calculation (numbers in parenthesis are **expected counts** calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

- It shows that **like_science_fiction** and **play_chess** are **correlated in the group**

Correlation Analysis (Numeric Data)

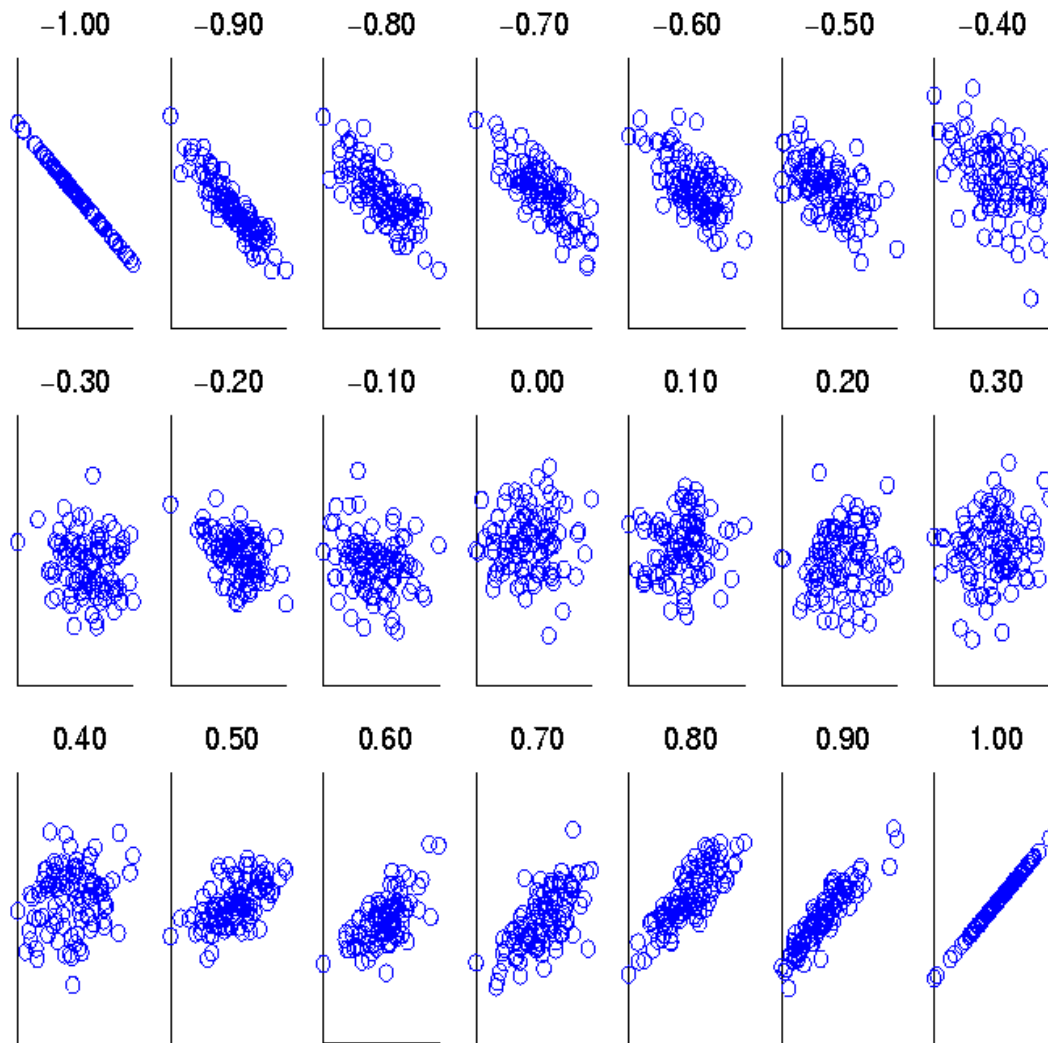
- Correlation coefficient (also called **Pearson's product moment coefficient**)

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective means of A and B , σ_A and σ_B are the respective standard deviation of A and B , and $\sum(a_i b_i)$ is the sum of the AB cross-product

- If $r_{A,B} > 0$, A and B are **positively correlated** (A 's values increase as B 's). The higher, the stronger correlation
- $r_{A,B} = 0$: independent; $r_{AB} < 0$: negatively correlated

Visually Evaluating Correlation



Scatter plots
showing the
similarity
from -1 to 1

Correlation

- Correlation measures the **linear relationship between objects**
- To compute correlation, we standardize data objects, A and B, and then **take their dot product**

$$a'_k = (a_k - \text{mean}(A)) / \text{std}(A)$$

$$b'_k = (b_k - \text{mean}(B)) / \text{std}(B)$$

$$\text{correlation}(A, B) = A' \bullet B'$$

Covariance (Numeric Data)

- Covariance is similar to correlation

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

Correlation coefficient: $r_{A,B} = \frac{Cov(A, B)}{\sigma_A \sigma_B}$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective mean or **expected values** of A and B , σ_A and σ_B are the respective standard deviation of A and B

- Positive covariance:** If $Cov_{A,B} > 0$, then A and B both tend to be larger than their expected values
- Negative covariance:** If $Cov_{A,B} < 0$ then if A is larger than its expected value, B is likely to be smaller than its expected value
- Independence:** $Cov_{A,B} = 0$ but the converse is not true:
 - Some pairs of random variables may have a covariance of 0 but are not independent. Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence

Covariance: An Example

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

- It can be simplified in computation as

$$Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

- Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14).
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
 - $E(A) = (2 + 3 + 5 + 4 + 6) / 5 = 20 / 5 = 4$
 - $E(B) = (5 + 8 + 10 + 11 + 14) / 5 = 48 / 5 = 9.6$
 - $Cov(A, B) = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$
- Thus, A and B rise together since $Cov(A, B) > 0$

Rangkuman

1. **Data quality**: accuracy, completeness, consistency, timeliness, believability, interpretability
2. **Data cleaning**: e.g. missing/noisy values, outliers
3. **Data reduction**
 - Dimensionality reduction
 - Numerosity reduction
4. **Data transformation** and data discretization
 - Normalization
5. **Data integration** from multiple sources:
 - Entity identification problem
 - Remove redundancies
 - Detect inconsistencies

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